

Convergence Analysis of CRR and LeisenReimer Binomial Models to Black-Scholes

03/04/2026

Introduction

This report presents a detailed analysis of the convergence properties of two prominent binomial option pricing models: the Cox-Ross-Rubinstein (CRR) model and the Leisen-Reimer (LR) model, benchmarked against the Black-Scholes (BS) analytical solution. Their accuracy is highly dependent on the number of steps (N) used in the binomial tree. The primary objective of this study is to examine how quickly and smoothly the prices generated by CRR and LR models converge to the Black-Scholes price under various market conditions, including different moneyness levels (At-The-Money, In-The-Money, Out-Of-The-Money) and volatility regimes. Binomial models, first proposed by Cox, Ross, and Rubinstein, offer a discrete-time alternative that can approximate the Black-Scholes price as the number of steps approaches infinity. The Leisen-Reimer model is an enhancement designed to improve the convergence speed and accuracy of binomial trees, particularly by adjusting the probabilities to match the Black-Scholes cumulative normal probabilities more closely.

Methodology

To conduct this convergence analysis, the following models were implemented in Excel via VBA.

To implement the CRR model and the LR I used the scripts from [VinegarHill-Finance Lab](#)

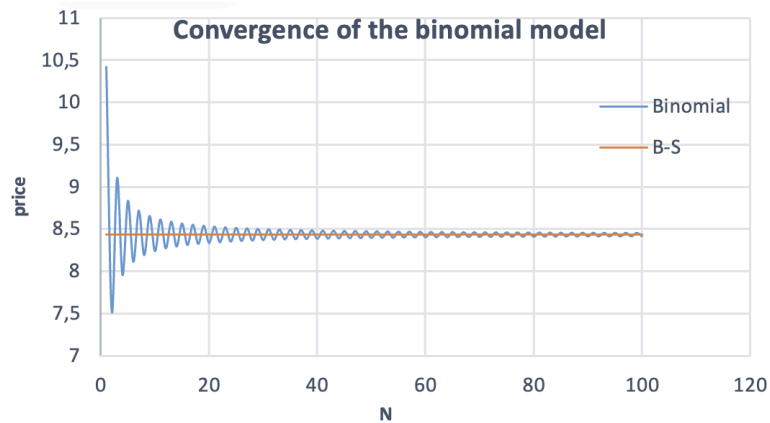
- Black-Scholes Model:** Used as the benchmark for European call option pricing. The formula for a European call option is: $C = Se^{-qt}N(d1) - Ke^{-rt}N(d2)$.
- Cox-Ross-Rubinstein (CRR) Binomial Model:** This model constructs a binomial tree where the stock price can either move up by a factor or down by a factor. The parameters are defined as: $u = e^{\sigma\sqrt{\Delta T}}$, $d = e^{-\sigma\sqrt{\Delta T}}$, $q = (e^{r\Delta T} - d)/(u - d)$ where $\Delta T = T/N$ is the time step and q is the risk-neutral probability of an upward movement. Option prices are calculated by working backward from the maturity date.
- Leisen-Reimer (LR) Binomial Model:** The LR model aims to improve convergence by adjusting the probabilities of up and down movements such that the first two moments of the terminal stock price distribution match those of the Black-Scholes model. This is achieved through the Peizer-Pratt inversion formula, which maps the Black-Scholes $d1$ and $d2$ values to binomial probabilities. The key innovation lies in the calculation of the risk-neutral probability and the up/down factors and using the Peizer-Pratt inversion function $h(z, N)$: $p = h(d2, N)$, $p' = h(d1, N)$, $u = e^{r\Delta T} p'/p$, $d = (e^{(r-q)\Delta T} - pu)/(1-p)$. The Peizer-Pratt inversion function is given by:

$$h(z, N) = 0.5 + \text{sign}(z) \cdot 0.5 \cdot \sqrt{1 - \exp \left[- \left(\frac{z}{N + \frac{1}{3} + \frac{0.1}{N+1}} \right)^2 \cdot \left(N + \frac{1}{6} \right) \right]}$$

The LR model typically requires an odd number of steps (N) for optimal performance.

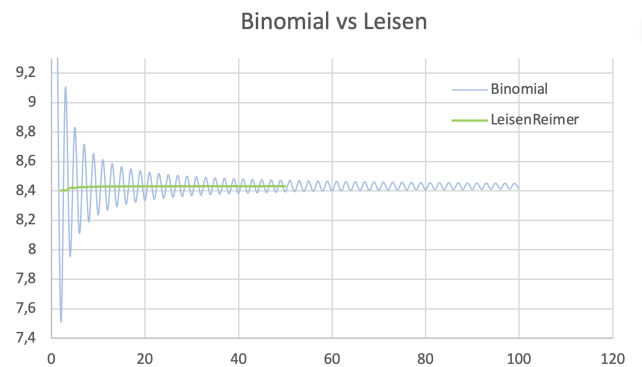
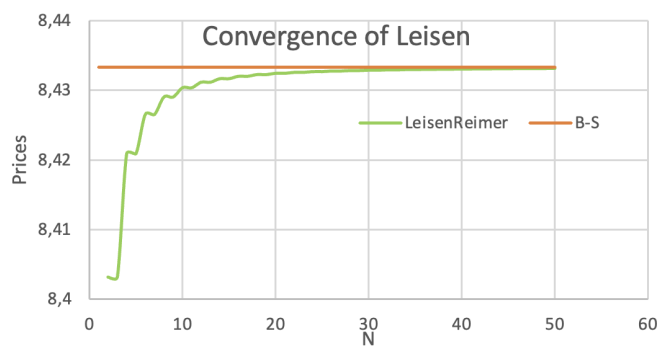
ATM OPTION

CRR



The analysis shows that as the number of steps N increases, the binomial price does not converge linearly towards the Black-Scholes value, but exhibits marked oscillations defined as "sawtooth convergence". This phenomenon occurs because the strike K changes relative position with respect to the final nodes of the tree with each increment of N .

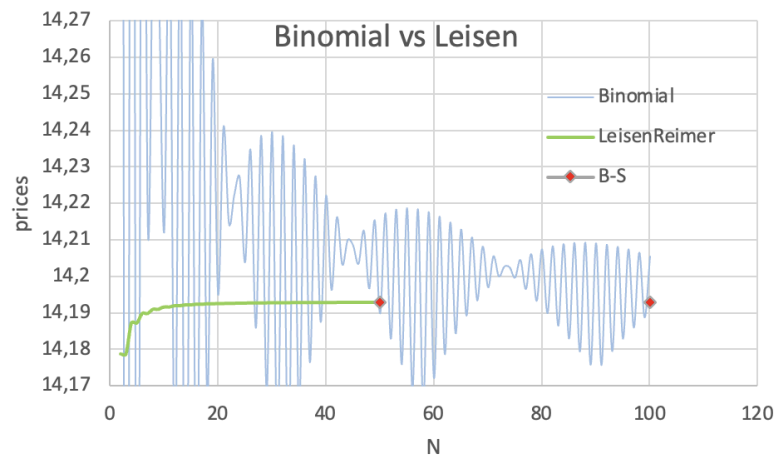
LR



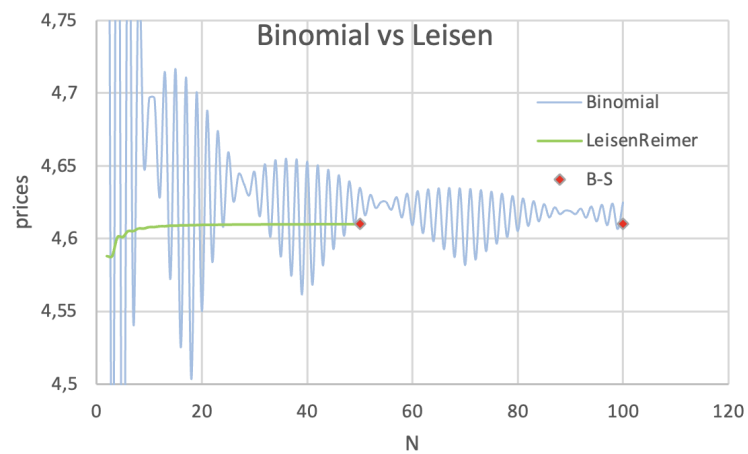
The Leisen-Reimer model demonstrates significantly faster and smoother convergence, approaching the Black-Scholes price with fewer steps and minimal oscillations

ITM and OTM OPTIONS

ITM



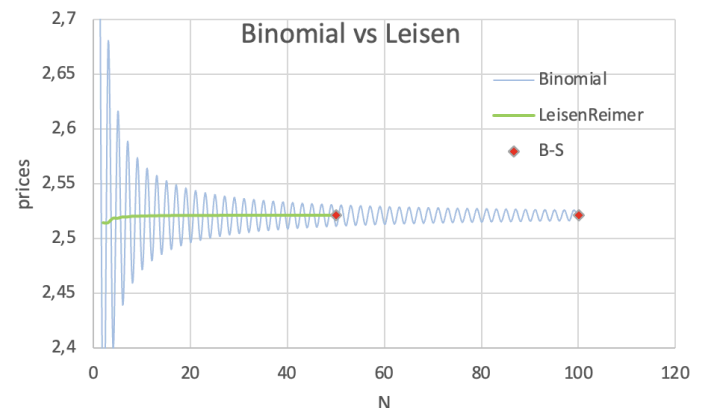
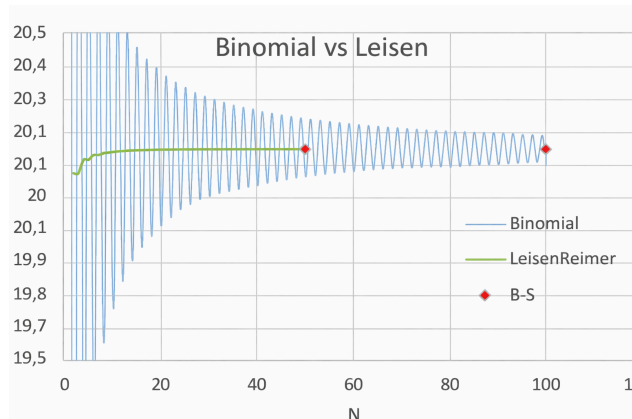
OTM



Similar to the ATM case, the CRR model exhibits oscillations around the Black-Scholes price, though the magnitude of these oscillations might vary. The LR model again shows a rapid and monotonic convergence towards the Black Scholes value. This consistent behavior across different moneyness levels highlights the effectiveness of the LR model's design in mitigating the oscillatory nature of standard binomial models. The ability of LR to converge quickly is particularly valuable in practical applications where computational efficiency is crucial

HIGH AND LOW VOLATILITY

RESPECTIVELY



When volatility is high, the option price is generally higher, and the range of possible stock prices at maturity is wider. In this scenario, the CRR model's oscillations tend to be more pronounced, making its convergence slower and less stable. The larger price movements associated with high volatility exacerbate the discrete nature of the CRR tree. Conversely, the Leisen-Reimer model continues to exhibit strong convergence properties, adapting well to the increased volatility

In a low volatility environment, the stock price movements are more constrained, leading to lower option prices. For the CRR model, the oscillations might appear less dramatic in absolute terms compared to high volatility, but the relative error can still be significant for a small number

of steps. The Leisen-Reimer model once again demonstrates superior convergence, quickly settling on the Black-Scholes price with minimal deviation.

these indicates that the LR model's enhancements are effective across the entire spectrum of volatility levels

CONCLUSIONS

The analysis clearly demonstrates the superior convergence characteristics of the Leisen-Reimer (LR) binomial model compared to the Cox-Ross-Rubinstein (CRR) model when benchmarked against the Black-Scholes analytical solution for European call options. The CRR model consistently exhibits oscillatory convergence, meaning its price estimates fluctuate around the true Black-Scholes value. This oscillatory behavior is a known limitation of standard binomial models and can lead to significant pricing errors, especially when the number of steps in the tree is small. The Leisen-Reimer model consistently achieves faster and smoother convergence across all tested scenarios (ATM, ITM, OTM, high volatility, and low volatility). By centering the binomial tree more effectively, the LR model minimizes the discrete approximation errors inherent in binomial methods, leading to more accurate and stable price estimates with fewer steps

From a practical perspective, the rapid and monotonic convergence of the LeisenReimer model makes it highly advantageous. In real-world applications, computational efficiency is often a critical concern. The ability to achieve accurate option prices with a smaller number of steps translates directly into reduced computation time, which is particularly beneficial for pricing complex options or performing large-scale simulations.